Bihar Engineering University, Patna
VIIIth Semester Examination - 2023

Course: B.Tech. *Time: 03 Hours*
Code: 105816 *Subject: Embedded System* *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option of the following (Any seven question only): **[2 x 7 = 14]**

(a) An embedded system is a/an _____ system.
(i) electronic (ii) mechanical
(iii) electro-mechanical (iv) either (i) or (iii)

(b) Which of the following is not the load-store type of ARM instruction?
(i) LDRH (ii) STRB
(iii) TST (iv) ADR

(c) Polling is seen in _____ I/O.
(i) normal (ii) busy-wait
(iii) interrupted (iv) None of the above.

(d) _____ is not a RISC architecture.
(i) ARM (ii) MIPS
(iii) SHARC (iv) Power PC

(e) Stage that reads program data from code memory to instruction buffer queue is known as
(i) Execution stage (ii) Decode stage
(iii) Address stage (iv) Fetch stage

(f) Which memory storage is widely used in PCs and Embedded System?
(i) EEPROM (ii) Flash memory
(iii) SRAM (iv) DRAM

(g) The internal RAM memory of 8051 is
(i) 32 bytes (ii) 128 bytes
(iii) 64 bytes (iv) 256 bytes

(h) Which design can be used to reduce the energy consumption of the embedded system?
(i) Simulator (ii) Compiler
(iii) Emulator (iv) Debugger

(i) Which of the following is true about the Direct Memory Access (DMA) technique in an embedded system?
(i) It is used for analog-to-digital conversion.
(ii) It is used for controlling physical system.
(iii) It is used for transferring data between memory and peripherals
(iv) It is used for event-driven system.

(j) Processor must accept and process frame before next frame arrives, typically called
(i) hard real-time systems. (ii) Real-data constraints.
(iii) Real-time constraints. (iv) Soft real-time systems.

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Q.2 (a) What is an embedded system? Comment on the embedded system characteristics. **[5]**

(b) What is the role of microprocessor or microcontroller in embedded system design? List down other technologies usable for embedded design. **[5]**

(c) Briefly explain the embedded system design process. **[4]**

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Q.3	(a)	Define requirements in perspective of embedded system design process. Explain with proper categorization of the requirements.	[8]
	(b)	Take an example application "GPS Moving Map". Create a requirement form which well describes the requirements of this application, for its design.	[6]
Q.4	(a)	Explain the all steps of compilation techniques with appropriate examples.	[6]
	(b)	Discuss the debugging techniques in short.	[4]
	(c)	Discuss the different types of code optimization used for assembly code compilation.	[4]
Q.5	(a)	How do we interface memory chip with CPU? Also explain the multichip memory interfacing.	[7]
	(b)	Differentiate between I/O instructions and memory mapped instructions. Why do most CPU architectures use memory-mapped I/O? Explain.	[7]
Q.6	(a)	Explain the priority inheritance protocol.	[7]
	(b)	Explain memory management in task control Block-model.	[7]
Q.7	(a)	Discuss the instruction set available in ARM processor with example.	[7]
	(b)	Discuss about the special features of SHARC processor as compared with ARM processor.	[7]
Q.8	(a)	Why do you require a limit on the minimum size of Ethernet frame?	[7]
	(b)	What are the different types of cabling supported by Ethernet standard?	[7]

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) ARM Processor
(b) SHARC Processor
(c) Basic compilation techniques
(d) Component interfacing

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